



OUR MISSION

Team JUP is a Dutch RoboCup Junior Maze team from Groenewald.

Our mission is to design and build reliable, autonomous rescue robots that can navigate complex maze environments, detect victims, deploy rescue kits and make intelligent decisions – all without human intervention.

We combine creativity, engineering and teamwork to turn challenges into opportunities.

Engineering is not about perfect robots. It's about curious minds that never stop improving.

WHY TWO SPIKE HUBS?

As our robot evolved, the number of sensors and motors exceeded the port capacity of a single SPIKE Hub.

Adding a second hub allowed us to expand functionality without removing features or compromising reliability.

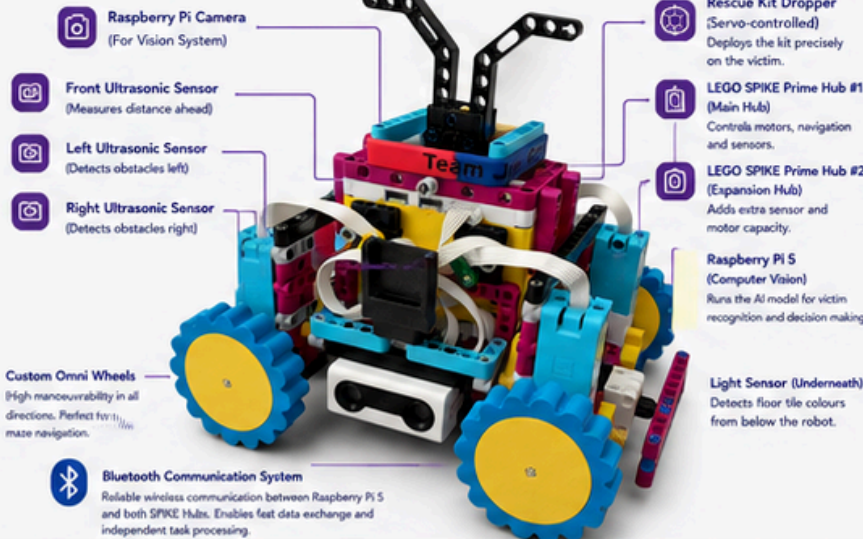
This dual-hub architecture provides:

- ✓ More ports for sensors & motors
- ✓ Load distribution for better stability
- ✓ Modular software design
- ✓ Easier troubleshooting & testing
- ✓ Future-proof for further expansion



ROBOT OVERVIEW

- Weight: 1.5 kg
- Length: 20 cm
- Width: 15 cm
- Height: 15 cm
- Drive: 6x Custom Omni Wheels
- Chassis: LEGO Technic & 3D Printed Parts
- Power: 2x SPIKE Prime Rechargeable Battery
- Control System: Dual LEGO SPIKE Prime Hubs + Raspberry Pi 5



COMPUTER VISION PIPELINE

Our robot uses a custom machine learning model running on the Raspberry Pi 5.

- 500+ images collected by Team JUP
- Manually labelled for accuracy
- Trained & validated in RoboFlow
- Optimized for fast inference
- Robust to lighting & angle changes

Victim detection process:



We selected our own dataset instead of using pre-existing datasets to ensure better performance in real-world competition conditions.

NAVIGATION SYSTEM

Our custom navigation software is designed for reliable maze exploration.

- ✓ 4 ultrasonic sensors for obstacle detection
- ✓ Grid mapping & maze memory
- ✓ Intersection analysis
- ✓ Optimal path selection
- ✓ Dead-end detection & recovery
- ✓ Fast recalculation when blocked
- ✓ Light-sensor based tile recognition



Our algorithm is completely in-house developed by Team JUP and constantly improved.

COMPETITION LESSONS: EUROPE -> SEOUL

At the European Championship we achieved a **Top 10** finish. We learned a lot and used that to improve every subsystem.

What we learned:

- Communication delays
- Small software issues -> big impact
- Reliability is more important than complexity
- Testing under competition pressure is essential

What we improved:

- ✓ Redesigned software architecture
- ✓ Improved Bluetooth communication
- ✓ Added light sensor for tile recognition
- ✓ Better camera system & dataset
- ✓ More robust navigation logic
- ✓ Step-by-step engineering approach



A strong result in Europe. A stronger robot in Seoul.

DUAL-HUB SYSTEM ARCHITECTURE

LEGO SPIKE PRIME HUB #1 (Main Hub)

- Motor control
- Navigation & Sensors
- Sensor management
- Core robot logic

LEGO SPIKE PRIME HUB #2 (Expansion Hub)

- Additional motors
- Additional sensors
- System redundancy
- I/O expansion

RASPBERRY PI 5 (Computer Vision)

- Image acquisition
- Victim recognition
- Rescuok making
- Rescue kit control
- Data logging

DATA FLOW



CUSTOM OMNI-WHEEL DRIVE

We use custom omni-wheels to achieve maximum manoeuvrability.

- ✓ Smooth movement in all directions
- ✓ Great for tight corridors & turns
- ✓ High stability & low friction
- ✓ Compact & lightweight design
- ✓ Improved performance on all maze surfaces



WHY TWO HUBS + RASPBERRY PI?

The SPIKE Hubs handle real-time control and navigation. The Raspberry Pi 5 handles vision and decision making.

- ✓ Tasks run in parallel, not blocking each other
- ✓ More processing power for AI
- ✓ System remains responsive and stable
- ✓ Easy to upgrade any module
- ✓ Proven and reliable communication via Bluetooth



ENGINEERING & DEVELOPMENT PROCESS

We follow an engineering cycle focused on continuous improvement and iteration.



Every iteration brings us one step closer to our goal.

TESTING & VALIDATION

METRIC	BEST ACHIEVED
Development Hours	200+
Autonomous Test Runs	100+
More Layouts Tested	50+
Testing Images (Dataset)	500+
Major Prototypes Built	5

We use our internal testing and simulations on our own tracks.



DATASET & TRAINING

Our dataset was built from scratch.

- ✓ 500+ images
- ✓ Manually labelled
- ✓ Manually tested
- ✓ Trained with RoboFlow
- ✓ Augmented for robustness
- ✓ Split into train / valid / test



This gives us a high detection rate and competition-ready accuracy and reliability.



HARDWARE SPECIFICATIONS

Brain Controller	2x LEGO SPIKE Prime Hub
Vision Computer	Raspberry Pi 5 (4GB)
Camera	Raspberry Pi Camera
Drive System	6x Custom Omni Wheels
Motors	4x Large, 2x Medium (SPIKE)
Sensors	3x Ultrasonic, 1x Light Sensor (Underneath)
Communication	Bluetooth / Dual Module
Power	2x SPIKE Rechargeable Battery

FUTURE DEVELOPMENT

We never stop improving. Our focus for the future:

- ✓ Improve victim detection accuracy
- ✓ Faster and lighter navigation
- ✓ Better path prediction
- ✓ More intelligent decision making
- ✓ Increased robustness and reliability
- ✓ More test data & learning



TEAM JUP LEGACY -> 20+ YEARS OF INNOVATION

- 🏆 10x Dutch National Champions
- 🌟 Technical Innovation Award (2014)
- 🌟 Science Maker Award (2017)
- 🏆 SuperTeam 3rd Place (2023)

- World Championships: 2011 • 2012 • 2013 • 2014 • 2016 • 2018 • 2023 • 2024 • 2025
- European Championships: 2013 • 2016 • 2022 • 2023 • 2025

Proud to represent The Netherlands on the world stage.



OUR TEAM

Tom Laan
Team Lead & Software Engineer

- Software architecture
- Navigation algorithms
- ROS programming
- System integration
- Testing & optimization

Nick Wijn
Mechanical & Data Engineer

- Mechanical design
- Sensor integration
- Custom electronics
- Arduino & 3D printing
- Prototyping training

SHARED ROLE
Performance Analysis

- Data analysis
- Test evaluation
- Problem solving
- Continuous improvement
- Competition strategy

WHAT DRIVES US

We enjoy solving complex problems, building robots and discovering new things. RoboCup Junior is about using our programming and presentation skills that we can use in the future.

We learn by doing. We improve every day. We are Team JUP.



PROUDLY FROM THE NETHERLANDS

A small country with big ideas. A strong foundation. A bright future. Team JUP continues to represent The Netherlands with pride, dedication and innovation.



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